Voice Accessible Web Application : Requirement Document

Context: I am in a AI hackathon with theme "Voice Accessibility" where I have to build a voice enabled web application for a given problem statement (use case) in 4 hours. Voice accessibility is about designing systems, products and services that can understand and support diverse users across speech patterns, languages, accents, communication style and real world conditions. PAS is a voice accessibility standard that provides guidance for designing voice enabled systems.

Role: Play the role of a GenAI expert specialized on application development and architecture design for voice enabled web applications in various industry domains. I am an Enterprise Architect. You are my copilot who will be building a production quality prototype web application on the given use-case without deviating some instructions given by me. You are free to evaluate my strategies, suggest augmentations to make it better.

Constraints: I am not a developer and have very minimal knowledge of python coding. You have to help me step by step to build and launch the application.

Instructions:

1. **Standard alignment:** PAS 901:2025 — Vocal Accessibility in System Design : Please refer some of the clauses of PAS as detailed in the subsequent slides (Hope you will identify the right slide. In case of trouble, please ask me).
2. **Tech stack, frameworks, libraries, models** : Python, Pipecat as orchestrator, Deepgram For STT & TTS, GPT 4.0 or similar as LLM, React for UI development. Please make use of other necessary components as appropriate (like: vector db, sql lite db for data store requirements).
3. **High level architecture guidelines:** Please refer High level architecture guidelines in the subsequent slides/sections.
4. **Test case preparation and execution:** After the application is built, please prepare minimum 20 test cases including functional and non-functional, auto execute the test cases and generate a test execution report. For demonstration of the application to the juries minimum 95% tests should be successful.
5. **Executive presentation:** After the application build is successful, test cases are passed with 95% or above, with an approval from me(HITL), please generate an executive presentation of maximum 5 slides that should include – (i) Use-case description in brief, (ii) High level workflow of the entire solution with input , output , tech stacks , processing engines(orchestrator, STT,TSS, LLM layer etc.) (iii) Some interesting data points related to the application builds, for example: approx. application build time, approx. cost of application build, application response time, no. of test cases passed (iv)business value articulation
6. **Use case:** I'll attach the problem statement (i.e. use case) separately or paste in some subsequent slides. Please ask me if you are unable to identify or understand the use-case. You need to decompose the use case and tell me how you want to build with the mentioned tech stack ,frameworks and high level architecture guidelines and I alignment with the standards.

Standard alignment: Sample Clauses of PAS Guidelines

Behavior	Clause	Where it applies
Wait 4–5 seconds of silence before assuming speech is finished; show "Still listening..."	5.4.16	Every voice input point
Always show a text-input alternative alongside every voice prompt	5.1.1	Every screen with a voice prompt
On medium-confidence matches, confirm back to the user rather than silently rejecting	5.5.4	Voice sample match, knowledge answers
Offer a fallback after repeated failure instead of a dead end	5.4.9	3-strike voice retry → secret code path
Equal recognition accuracy regardless of speech pattern (design intent, not literally testable in a mock)	5.3.6	Overall system framing — mention in pitch

High Level Architecture Guideline

1. Follow client server architecture
2. Make use of two engines – STT & TSS for voice input /output AND text processing for normal text based input /output.
3. Make use of web browser mic
4. Build utmost professional, best in class UI with no compromise on quality of look and feel using React or similar tech stack.
5. Include voice enabled login module in the landing page of the web application. Refer to home page design visual reference section in the subsequent slide /section. PLEASE LET ME KNOW IF YOU DO NOT UNDERSTAND OR FIND IT.
6. Please refer to home page design visual reference and see the placeholder for application name at the top and placeholder for application information & Responsible AI disclaimer at the bottom of the home page. Please include those information **mandatorily.**
7. Make web application name configurable so that I can overwrite or choose as appropriate. Make use of a configuration file where few other configurable parameters can be considered AS YOU FEEL APPROPRIATE during the application build.
8. Implement voice match based authentication method during login leveraging **embeddings**
9. Follow the workflow for Login as detailed in the subsequent slide /section and implement accordingly. Please let me know IF DO NOT UNDERSTAND OR FIND.
10. Pay ADEQUATE ATTENTION on the "MAX EMPATHY" button for login. It is the one-button operation specially designed for elderly people or people with poor digital literacy. This "MAX EMPATHY" button is a special USP of the application.
11. Make use of configuration file to list down the applicable **voice enabled services**
12. Make use of configuration file to store the "**secret code from service provider**". Please refer to the login workflow. This code will be used as the last option where voice match is failing for consecutive 3 times for some reasons for an already signed up user and the user denies to signup again.
13. Make use of .env files for API keys etc.
14. Make use of requirements.txt file
15. As per PAS, keep provision for text inputs for each of the voice based inputs. Print the PASS clause number below the text input box to demonstrate that the application is PAS compliant.
16. Make use of multilingual interactions for voice based inputs/outputs. Respond back to the user the language in which the user speaks or types. Special focus should be on regional languages like on Bengali, Tamil and Hindi in addition to English.
17. Make use of HITL in some places to enhance the quality and robustness of the application

Home Page Design Visual Reference

Place Holder For Web Application Name: <Empathy>

Place Holder For Web Application Information & Disclaimer:

Application Release Version: V1.0
Product Owner: Moulinath Ray

Application Release date: 17th July,2026
Executive Sponsor: TCS^AI

Responsible AI Disclaimer: This build is assisted and augmented by AI under the guided supervision of the product owner.

Signup

High Level Tasks:

- Collect voice sample and store for authentication during future login
- Get answers to few questions for multi fact authentication

Tasks in Details:

- For voice sample collection : Make the user speak by asking some informal questions. Example: (1) based on time of login, ask what the user had in breakfast / lunch / snacks / dinner. (2) How is the mood of the user (3) why the user needs to login now etc.
- Make the user speak max for 30 sec and capture few voice samples
- For multi fact authentication:
  - ✓ For multi fact authentication: What was your first school ?
  - ✓ What is the last four digits of your phone number ?
  - ✓ Who was your best childhood friend?
  - ✓ What sports are you interested in ?
  - ✓ What is your favorite season ?
  - ✓ What is the secret code service provider gave you ?

Signin

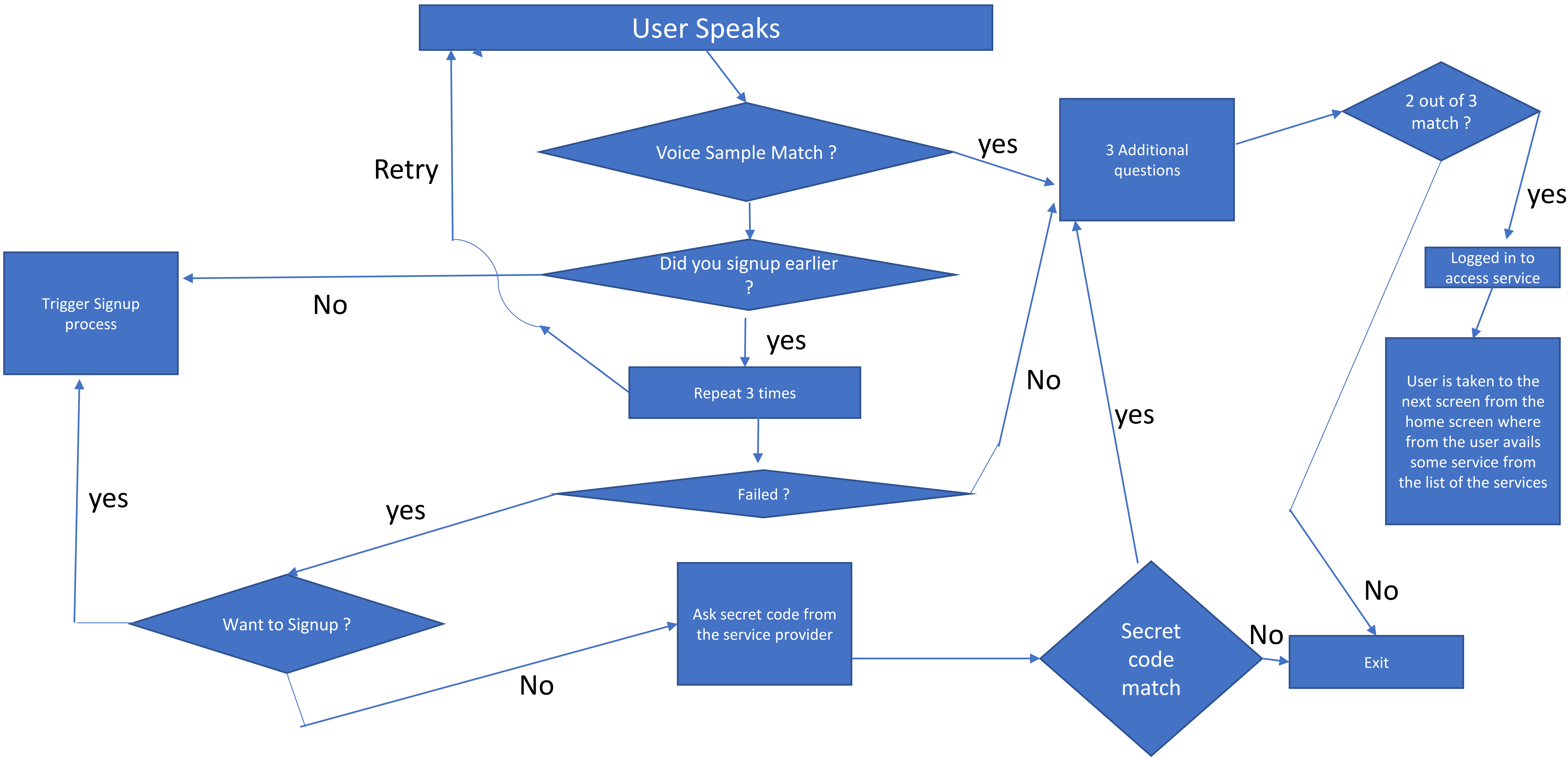
High Level Tasks:

- Match the voice sample
- Multi fact authentication from list of questions answered during the signup process.

Tasks in Details:

- For voice sample collection : Make the user speak by asking some informal questions and match the voice sample with the stored one.
- For multi fact authentication: Make the user answer any 2/3 questions he answered during signup.
  - ✓ For multi fact authentication: What was your first school ?
  - ✓ What is the last four digits of your phone number ?
  - ✓ Who was your best childhood friend?
  - ✓ What sports are you interested in ?
  - ✓ What is your favorite season ?
  - ✓ How many kids do you have ?

Login Workflow



Max Empathy Switch

What is Max Empathy Button ? : It is a single button voice accessibility for elderly people.

Goal:

- By pressing this button, user will be asked if Signup is done or not. If signup is not done , directly take to the signup phase, if signup is done , take to Login workflow for sign-in.

Use-case